NUMBER SYSTEM

COMPUTER FUNDAMENTALS

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GRADING SYSTEM

FINAL GRADE

30%

PRELIM

30%

MIDTERM

40%

ENDTERM

TERM GRADE

50%

WEEKLY

50%

EXAM

100%

LABORATORY

PENALTY

0

ABSENT

3

45-min LATE

7 max

15-min LATE

TOPIC OUTLINE

Number System

Number System Conversion

Adding Number System

NUMBER SYSTEM

Number System	Base/Radix	Digits
Decimal	**10**	**0 to 9**
Binary	**2**	**0 and 1**
Octal	**8**	**0 to 7**
Hexadecimal	**16**	**0 to 9** **A to F**

TABLE

Decimal	Binary	Octal	Hexadecimal
0	0000	0	0
1	0001	1	1
2	0010	2	2
3	0011	3	3
4	0100	4	4
5	0101	5	5
6	0110	6	6
7	0111	7	7
8	1000	10	8
9	1001	11	9

TABLE

Decimal	Binary	Octal	Hexadecimal
10	1010	12	A
11	1011	13	B
12	1100	14	C
13	1101	15	D
14	1110	16	E
15	1111	17	F

NUMBER SYSTEM CONVERSION

Successive **division** by the radix until the dividend reaches **zero**

$$\frac{\text{DEC}}{\text{RADIX/BASE}}$$

Least Significant Bit

LSB

MSB

Most Significant Bit

Convert 45_{10} to BIN

Q	R

Q	R

Convert 328_{10} to BIN

Q	R

Q	R

Convert 45_{10} to OCT

Q	R

Q	R

Convert 328_{10} to OCT

Q	R

Q	R

Convert 45_{10} to HEX

Q	R

Q	R

Convert 328_{10} to HEX

Q	R

Q	R

ANY NUMBER SYSTEM-TO-DECIMAL

Each digit is multiplied by its **positional notation**

MSB **NUMBER** LSB

$\times$ **base**^{0}

$\times$ **base**^{1}

$\times$ **base**^{2}

$\times$ **base**^{n}

$(+)$

Convert 101101_2 to DEC

Convert $\mathbf{101001000_2}$ to DEC

Convert 55_8 to DEC

Convert $\mathbf{510_8}$ to DEC

Convert $\mathbf{2D_{16}}$ to DEC

Convert $\mathbf{148_{16}}$ to DEC

Convert 1011011101_2 to OCT

1 Octal digit is equivalent to **3 Bits**

1011011101_2

Convert 1011011101_2 to HEX

1 Hex digit is equivalent to **4 Bits**

1011011101_2

Convert $\mathbf{742_8}$ to HEX

Convert $\mathbf{1B6_{16}}$ to OCT

ADDING NUMBER SYSTEM

Decimal Number System

0, 1, 2, 3, 4, 5, 6, 7, 8, **9**, carry 1

Octal Number System

0, 1, 2, 3, 4, 5, 6, **7**, carry 1

Binary Number System

0, **1**, carry 1

Hexadecimal Number System

0, 1, 2, 3, 4, 5, 6, 7, 8, 9,

A, B, C, D, E, **F**, carry 1

ADDING DECIMAL

$$
\begin{array}{r}
17_{10} \\
+ \quad 15_{10} \\
\hline
\end{array}
$$

ADDING OCTAL

$$
\begin{array}{r}
17_8 \\
+\quad 15_8 \\
\hline
\end{array}
$$

ADDING OCTAL

$$
\begin{array}{r}
354_8 \\
+\quad 724_8 \\
\hline
\end{array}
$$

ADDING HEX

$$
\begin{array}{r}
17_{16} \\
+ \quad 15_{16} \\
\hline
\end{array}
$$

ADDING HEX

$$\begin{array}{r} \mathbf{A5ED_{16}} \\ + \quad \mathbf{F73_{16}} \\ \hline \end{array}$$

ADDING BINARY

$$
\begin{array}{r}
1101_2 \\
+\quad 1001_2 \\
\hline
\end{array}
$$

ADDING BINARY

$$
\begin{array}{r}
110110_2 \\
+\quad 101111_2 \\
\hline
\end{array}
$$

LABORATORY